Shaheen Nabi

Research Engineer | LLM Post-Training, Reasoning & RL

devshaheen0@gmail.com | +91 6005081860 | GitHub | LinkedIn | Bengaluru, Karnataka

Profile

Research @ Indian Institute of Science | LLM Post-Training, RLVR, Mechanistic Interpretability

Research engineer focused on LLM post-training, reasoning and thinking models, and RL-based optimization (RLHF/RLVR) — evaluation, and efficient inference for open-weight LLMs. Currently conducting research at IISc on reinforcement learning, model distillation, and mechanistic interpretability, alongside hands-on freelance/consulting experience shipping production AI systems end to end.

Experience

Researcher

Indian Institute of Science (IISc), Bengaluru, Karnataka | Jun 2026 – Present · 1 mo

- Conducting research on reinforcement learning, model distillation, and mechanistic interpretability for reasoning and language models.

Machine Learning Engineer

Freelancer, Remote | Dec 2025 – Present

- Designing and implementing post-training systems for large language models, focused on reasoning models, RL, alignment, evaluation, and reproducible open-weight pipelines; built and open-sourced end-to-end training/eval systems reproducing modern post-training techniques from research papers.
- Implemented RL algorithms and training pipelines for LLMs — Policy Gradients, Actor-Critic, PPO, and GRPO-style optimization — within end-to-end post-training workflows covering SFT, preference optimization, reward design, RL, and evaluation.
- Trained, evaluated, and released open-weight reasoning models, checkpoints, datasets, and reproducible research artifacts; built training infrastructure for rollout generation, reward computation, policy optimization, checkpointing, and experiment tracking.
- Developed evaluation pipelines for mathematical reasoning, inference-time scaling, and self-consistency decoding, using Hugging Face Transformers, TRL, vLLM, and distributed GPU training environments.
- Conducted research-focused reproductions and ablations on reasoning emergence and RL dynamics in open models, releasing all systems and experiments publicly via GitHub and Hugging Face.

Artificial Intelligence Consultant

Freelance / Self-Employed, Jammu & Kashmir, India (Hybrid) | Oct 2024 – Oct 2025 · 1 yr 1 mo

- Consulted with US and Canadian clients to architect and deploy AI-powered products, delivering 10+ end-to-end solutions across recommendation systems, search, computer vision, and LLM applications.
- Designed and developed a multi-agent travel planning platform integrating multiple AI agents, vector search, external APIs, and automated workflows to generate personalized travel itineraries.
- Built retrieval and recommendation systems leveraging embeddings, vector databases, semantic search, and ranking pipelines to improve information discovery and user experience.
- Developed and integrated AI services with production backend systems through REST APIs, databases, cloud infrastructure, and microservice-based architectures.
- Led the design and deployment of computer vision-powered security monitoring solutions for commercial environments, utilizing object detection and real-time event processing pipelines.
- Worked across the full AI stack — data pipelines, model development, system architecture, deployment, mon-

itoring, and performance optimization — collaborating directly with founders, product teams, and engineers to translate business requirements into scalable AI systems.

Data Science Intern

iNeuron.ai, Bengaluru, Karnataka, India (Remote) | Jan 2025 – Jun 2025 · 6 mos

- Developed an object detection model using YOLOv5 to accurately identify and classify various crops/plants.
- Annotated 25,000 images, later open-sourced on Hugging Face, contributing to the broader research community.
- Designed and deployed a fully automated AI agents pipeline, streamlining post-detection research and insights for detected crops/plants.
- Trained the model on NVIDIA A100 GPUs, achieving high performance (~99% precision/recall, ~99% mAP@0.5) and optimizing for real-world deployment.
- Conducted extensive model testing and evaluation, ensuring robustness and accuracy across diverse agricultural environments.
- Deployed the solution using Jenkins, AWS ECR, and EC2, leveraging scalable infrastructure for real-time inference.
- Integrated an SMTP service enabling automated delivery of personalized, 1-minute AI-generated research reports directly to users' inboxes.

Data Scientist

NIELIT, Jammu & Kashmir, India (Hybrid) | Feb 2023 – Dec 2023 · 11 mos

- Worked with clients on end-to-end Data Science and Machine Learning projects, from data collection and analysis to model development and deployment.
- Built predictive models, recommendation systems, dashboards, and automated data workflows for real-world business use cases.
- Developed and integrated REST APIs to support scalable applications and machine learning solutions.
- Contributed to frontend development, creating user-friendly interfaces and data-driven web applications.
- Collaborated directly with clients to understand requirements, solve business problems, and deliver production-ready solutions.
- Worked extensively with Python, SQL, machine learning libraries, data visualization tools, and cloud-based services.
- Mentored students through hands-on projects, building practical experience in Data Science, Machine Learning, and software development.
- Delivered numerous freelance projects across analytics, AI, web development, and automation domains.

Selected Projects

Open Post-Training

Python, PyTorch, Hugging Face Transformers, TRL, vLLM, SGLang | 2026

- Open-source research engineering project for building the end-to-end post-training stack for reasoning language models — SFT, preference learning, RLHF/RLVR, evaluation, inference-time scaling, and scalable systems for frontier-level reasoning. Repo: github.com/shaheennabi/open-posttraining-system.
- Used as a personal research codebase to test post-training hypotheses: implements reusable SFT/preference-optimization/RL components, evaluates reasoning on benchmarks (MATH-500), and tracks RLVR-GRPO experiments with full eval harnesses.
- Ran a GRPO ablation on a base model (64.0% MATH-500 accuracy): a 5-step checkpoint dropped to 48.0% while a 50-step checkpoint recovered to 60.0%, with response length shrinking from 303 to ~220 tokens — evidence that sparse binary-correctness reward is a weak learning signal requiring more steps and better reward shaping.
- Organized into modular components — custom base-model/Qwen implementation, model-download pipeline,

reasoning-evaluation harness, generation utilities, and inference-time-scaling experiments — covering SFT, DPO/ORPO/SimPO preference optimization, RL-based post-training, and benchmark analysis.

Olmo-3 From Scratch

Python, PyTorch, Hugging Face | 2026

- Clean, from-scratch PyTorch implementation of AllenAI's Olmo-3 architecture (7B and 32B variants), designed as a minimal yet extensible foundation for post-training research — RLHF, preference optimization, and reasoning-focused systems. Repo: github.com/shaheennabi/Olmo3-from-scratch.
- Implemented Grouped Query Attention (GQA), YaRN-scaled RoPE for extended context (8K → 65K), sliding-window attention with periodic global-refresh layers, SwiGLU feed-forward blocks, and RMSNorm.
- Built an efficient autoregressive inference pipeline with a custom KV-cache module (unrotated key/value storage for RoPE compatibility), reducing cache memory via GQA (e.g. 32→8 KV heads on the 7B variant) while supporting bfloat16 inference.
- Structured as a modular, readable codebase (attention, MLP, RoPE, normalization, transformer assembly, KV cache, weight loading from Hugging Face) intended as a base for adding training and post-training pipelines on top of the architecture.

Technical Skills

Programming & Systems: Python, Linux (Ubuntu/Debian), Git, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA)

Deep Learning & LLMs: PyTorch, TensorFlow, Decoder-only Transformers, Autoregressive Modeling, Transformer Architectures, Attention Mechanisms, Causal Self-Attention, RoPE Positional Embeddings, Supervised Fine-Tuning (SFT), RLHF, PPO, Preference Optimization

LLM Systems & Inference: vLLM, SGLang, Efficient LLM Inference, Tokenization Pipelines, KV Cache, Decoding Strategies, Training Pipelines

Reinforcement Learning: PPO, Policy Gradients, Actor-Critic Methods, Generalized Advantage Estimation (GAE), MDPs, Q-Learning, SARSA, Reward Design

MLOps & Infrastructure: Docker, MLflow, DVC, Airflow, Jenkins, GitHub Actions, AWS (EC2, S3, ECR, SageMaker)

Distributed & Training Optimization: Mixed Precision Training, Gradient Accumulation, Training Stability, GPU-based Training Workflows

Maths: Linear Algebra, Calculus, Probability & Statistics, Hypothesis Testing, Optimization Fundamentals

Education

Bachelor of Computer Applications

Indira Gandhi National Open University | Information Technology, Machine Learning, Deep Learning, and AI Systems